

Liang Gao

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Education

- George Washington University**, MS in Data Science | GPA: 3.9/4.0 Jan 2023 – Dec 2024
- **Coursework:** Deep Learning, Machine Learning, NLP, Time Series Modeling, Data Visualization, Data Warehousing
- Tulane University**, MS in Finance | GPA: 3.7/4.0 Aug 2021 – Dec 2022
- **Coursework:** Financial Modeling, Fixed Income Analytics, Accounting Analytics
- Xi'an Jiaotong-Liverpool University | University of Liverpool** Sept 2017 – Jun 2021
- Bachelor of Arts – Business Management (2+2 Program)
- **Coursework:** Quantitative Methods, Financial Management, International Finance

Technical Skills

Programming: Python, R, SQL

Tools: Tableau, Git, Linux, Azure DevOps, HTML, Excel

Libraries: NumPy, Pandas, Scikit-learn, PyTorch, Matplotlib, Plotly, SimpleITK, OpenCV

Work Experience

- Data Science Intern** Mar 2025 – Current
- Siemens Healthineers, NJ
- Processed and analyzed 5,000+ CT/MRI images using NumPy, Pandas, and SimpleITK to enhance radiotherapy planning and improve clinical data quality for AI-driven research.
 - Developed and automated reproducible data pipelines for radiotherapy treatment planning, improving data consistency and processing efficiency.
 - Designed and implemented data visualization reports in Matplotlib and HTML to support quality assessment and cross-functional decision-making.
 - Maintained robust Git and Azure DevOps workflows to streamline collaboration and code reproducibility.
- Graduate Teaching Assistant** Sep 2024 – Dec 2024
- George Washington University, DC
- Mentored 40+ students in statistical learning and data science fundamentals.
 - Automated NLP exam grading by designing an auto-grader with Python, improving grading efficiency and consistency across cohorts.

Technical Projects Experience

- Time Series Modeling - Capstone Research Project | Python, PyTorch** Aug 2024 – Dec 2024
- Built and compared predictive models (ARIMA, LSTM, Transformer) to forecast operational metrics, driving data-informed strategy recommendations. Deployed models on AWS EC2 for scalable training and inference.
 - Published as first author in *Neural Computing and Applications* journal.
- Biomedical Image Segmentation with CNNs | Python, PyTorch** Jan 2024 – May 2024
- Analyzed MRI data and developed a data pre-processing pipeline for AI model training.
 - Implemented U-Net CNN for multi-organ MRI segmentation, optimizing model structure and hyperparameters to achieve a Dice score of 0.93.
- Temperature Forecasting with Regression and Time-Series Models | Python** Jan 2024 – May 2024
- Conducted Exploratory Data Analysis on 17,545 observations and forecasted Germany's temperature using Multiple Linear Regression and SARIMA models.
 - Applied hypothesis testing to validate statistical significance, achieving an adjusted R^2 of 0.97.
- AI Book Recommendation System with Transformer | Python, PyTorch** Aug 2023 – Dec 2023
- Processed and analyzed 50,000+ book reviews using Pandas, Scikit-learn, and NLP techniques (NLTK, SpaCy).
 - Developed a recommendation model leveraging pre-trained BERT embeddings; deployed via Streamlit on AWS for real-time inference.
- Analyzing Robbery Crime in Washington D.C. | R** Jan 2023 - May 2023
- Analyzed 8,343 robbery incidents in Washington D.C. to identify socio-environmental drivers of crime, producing insights with potential policy implications.
 - Applied machine learning to interpret human behavior, supporting data-driven policy recommendations.